

Capstone Project I: Mutual Fund Analytics

Day 2 Report — Data Cleaning + SQL Database Design

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Git Repository: github.com/anveshtammineni/mutual-fund-analytics
Commit Message: Day 2: Cleaned data + SQLite DB loaded

1. Data Cleaning Implementation

A robust Python ETL data cleaning script (`data_cleaning.py`) was developed using Pandas. The script processed all 10 datasets, loaded from raw storage, and written to processed directories under the following constraints:

- **NAV History Expansion:** Parsed dates, dropped duplicates, and verified $\text{NAV} > 0$. By grouping by scheme and reindexing to daily frequency from minimum to maximum dates, we forward-filled (`ffill`) NAVs for weekends and holidays. This expanded ``02_nav_history.csv`` from 46,000 to 64,320 rows.
- **Transaction Standardization:** Standardized transactions into three types (``SIP``, ``Lumpsum``, ``Redemption``), validated transaction amounts > 0 , set date formats, validated KYC status tags to a strict enum (``Verified`/`Pending``), and removed duplicates.
- **Performance Validation:** Coerced return figures to numeric values, used category-based medians to impute minor omissions, and enforced the expense ratio bounds between 0.1% and 2.5%.
- **Standard Cleaning (All Other Datasets):** Standardized date formats, stripped leading/trailing spaces, deduplicated records, and outputted all 10 cleaned CSV files in `data/processed/`.

2. SQL Star Schema Database Design

A conformed relational Star Schema was modeled to optimize queries. A Python script (db_loader.py) created the SQLite database bluestock_mf.db, applied foreign key constraints (PRAGMA foreign_keys = ON), and loaded data. The schema design consists of the following conformed tables:

- **dim_fund**: Primary dimension table containing scheme features, benchmark codes, and limits.
- **dim_date**: Shared date dimension, generated programmatically containing 1,608 unique dates, year, month, day, quarter, weekday, and weekend flag.
- **fact_nav**: Historical Net Asset Values over time (linked to fund and date dimensions).
- **fact_transactions**: Investor execution tracking, containing demographic details and amount values.
- **fact_performance**: Returns, ratings, volatility standard deviation, and drawdowns.
- **fact_aum**: AMC Assets Under Management over time.
- **Additional facts**: Tables for SIP Inflows, Category Net Inflows, Folios, Sector Holdings, and Indices.

3. Ingestion Row-Count Verification

The database loader verified that every row from the cleaned source CSVs was loaded correctly. The comparison between CSV row counts and SQLite records shows complete parity:

SQLite Table Name	Source Cleaned CSV File	Record Count	Status
dim_fund	01_fund_master.csv	40	Match [OK]
fact_nav	02_nav_history.csv	64,320	Match [OK]
fact_transactions	08_investor_transactions.csv	32,778	Match [OK]
fact_performance	07_scheme_performance.csv	40	Match [OK]
fact_aum	03_aum_by_fund_house.csv	90	Match [OK]
fact_sip_inflows	04_monthly_sip_inflows.csv	48	Match [OK]
fact_category_inflows	05_category_inflows.csv	144	Match [OK]
fact_industry_folios	06_industry_folio_count.csv	21	Match [OK]
fact_portfolio_holdings	09_portfolio_holdings.csv	322	Match [OK]
fact_benchmark_indices	10_benchmark_indices.csv	8,050	Match [OK]

4. Analytical Queries & Reports Summary

Ten analytical SQL queries were coded in queries.sql and executed to generate metrics on the loaded database. Key analytical insights include:

- **Scheme AUM Leaderboard:** Mirae Asset Emerging Bluechip Fund is the largest equity fund in the database (AUM: 49,046 Cr), followed closely by Kotak Emerging Equity Fund (47,469 Cr).
- **SIP YoY growth:** Growth percentage is missing for the first 12 months as expected. YoY calculations show a steady rise in overall industry inflows over time.
- **Geographical Revenue distribution:** Punjab (31.57 Cr) and Tamil Nadu (31.51 Cr) contributed the highest transaction volumes in Rupees, followed closely by Madhya Pradesh (30.83 Cr).
- **Low Expense Funds:** Nippon India Gilt Securities is the cheapest scheme (expense ratio: 0.55%), followed by HDFC Short Term Debt (0.56%).
- **Sector Holdings:** Consumer Goods (avg weight 14.18%) and IT (11.39%) are the top heavily allocated sectors. Axis Bank (16,325 Cr total value) is the most heavily held equity stock.

5. Deliverables Checklist

Deliverable	Location / Details	Status
10 Cleaned CSVs	data/processed/ (duplicates removed, forward-filled NAV)	COMPLETED
SQLite Database	bluestock_mf.db (conformed star schema model)	COMPLETED
Schema DDL File	schema.sql (auto-exported by loader script)	COMPLETED
Analytical Queries	queries.sql (10 optimized SQL statements)	COMPLETED
Data Dictionary	data_dictionary.md (table and column reference guide)	COMPLETED
Walkthrough Logs	walkthrough.md (logs covering Day 1 + Day 2)	COMPLETED
PDF Report	reports/Day_2_Database_Design_Report.pdf	COMPLETED

End of Day 2 Report